

E-Commerce Sales Analysis – SQL Case Study

1. Case Study Summary

This case study focuses on analyzing an e-commerce orders dataset using SQL. The objective is to understand sales performance, customer purchasing behavior, and product demand trends to support strategic business decisions.

2. Problem Statement

The company wants to gain insights from its sales data to answer key business questions such as: How much revenue is being generated? Which products are selling the most? Who are the most valuable customers? And how many customers are returning for repeat purchases?

3. Key Performance Indicators (KPIs)

- Total Revenue from all orders
- Monthly Revenue trends
- Top 5 most sold product categories
- Top 3 high-value customers by total spending
- Repeat customer count

4. Key Insights

- Total revenue provides a clear overview of the business's overall sales performance.
- Monthly revenue analysis helps identify growth patterns and seasonal variations.
- Some product categories significantly outperform others in terms of sales volume.
- A small group of customers contributes a large portion of total revenue.
- Repeat customers indicate strong customer loyalty and retention.

5. Graphs and Visualizations (Optional)

Visualizations such as bar charts and line graphs can be created using tools like Power BI, Excel, or Tableau to represent monthly revenue trends, product-wise sales, and customer contribution more effectively.